How large is the US data-centre boom?

Macro and macro-financial ballparks

September 2026

Three questions for the Moody's roundtable, with a first pass at the arithmetic. Investment through 2027 is Brandsaas, Garcia, Kurtzman, Nichols and Zytek (FEDS 2025-109), Dodge vintage September 2025. Census structures through July 2026 are the live check. Electricity is LBNL's 2025 Update. Financing numbers are BIS, BoE, FSB and the 2025 IG calendar. Anything after 2027, and the 2035 electricity share, is a mechanical hold of a 2026–27 rate.

1 Questions

I_t^{DC} is US data-centre GFCF (structures plus installed IT). I_t is a US investment aggregate. E_t^{DC} is data-centre electricity consumption; E_t is US end-use electricity. F_t is the funds this investment absorbs: retained earnings spent out of cash and bills, plus bonds, private credit, bank loans and ABS/CMBS.

1. Share of investment. What is $s_t^I \equiv I_t^{DC}/I_t$ in 2030 and 2035, against 2023? Report both gross private domestic investment (GPDI) and private non-residential fixed investment (PNFI).

2. Share of electricity, and the price. What is $s_t^E \equiv E_t^{DC}/E_t$ in 2030 and 2035, against 2023? Holding other demand fixed,
$$\frac{\Delta P}{P} \simeq \frac{1}{\varepsilon_S} \frac{E_T^{DC} - E_{2023}^{DC}}{E_{2023}}. \tag{1}$$
Section 6 puts a range on ε_S instead of setting it to 1.

3. Funds-market absorption. How large is F_t next to IG credit, private credit and bank C&I, and next to the saving pool that prices r^*? Can that move interest rates, and in which market?

2 Assumptions

Table 1: Working assumptions

Object	Assumption
I^{DC} through 2027	Fed Table 2, Q4 annual rate, Dodge vintage September 2025. 30% structures, 70% equipment; 80% of equipment imported, so 44% of I^{DC} is domestic.
Live structures check	Census C30, July 2026: \$75.2 bn SAAR (+57% y/y).
US investment, 2025	GPDI \$5.46 tn (BEA Table 5.2.5). PNFI \$4.25 tn (FRED PNFIA). Grown at 2% real.
2030 and 2035 I^{DC}	Hold a 2026–27 Fed rate. Low: 2027 0.25× inflow (\$357 bn). Central: 2026 1× (\$464 bn). High: 2027 1× (\$612 bn).
Electricity, 2023–24	176 TWh (4.4% of 4,011 TWh) and 192 TWh (4.7% of 4,110).
Electricity, 2030	LBNL 2025 Update: 649 TWh, 11.8% of the NERC 2025 LTRA total (range 9.5–15.3%; 521–843 TWh). Data centres are 33% of 2024–30 load growth.
Electricity, 2035	Hold the 2030 share, or keep the 2023–30 drift to about 17%.
ε_S	0.3 short-run / local peak; 1.0 medium-run national generation; 2.5 long-run renewables.
External-finance share	30–50% of I^{DC}.
Funds-supply elasticity	$\eta = 1$; 0.5 if saving is less elastic.

3 The Fed nowcast is a year old

FEDS 2025-109 has three objects, and only the first two are pinned by data.

- History, matched to a NIPA imputation (Table 2, Q4 annual rate): \$10 bn in 2023, \$58 bn in 2024, \$179 bn in 2025.
- Nowcast from the existing Dodge plans, no assumed future inflows: mean \$370 bn by 2026Q2, 5th–95th \$240–450 bn. Most of the width is which mega-projects get abandoned (historical $\lambda = 0.33$).
- Forecast through 2027, which needs an inflow assumption: \$357 bn if new plans run at one-quarter of the 2024–25 pace, \$612 bn if they continue, \$921 bn if they double. After about a year the current planning stock no longer sets the level.

The Dodge extract is September 2025. We are now in the year where the nowcast runs out and the inflow assumption takes over. Census structures kept rising: \$50.7 bn SAAR in April 2026, \$70.8 bn in June, \$75.2 bn in July (+57% y/y). At the paper's 30/70 split, \$75 bn of structures is

about \$250 bn all-in, under the \$370 bn 2026Q2 mean. Equipment is more backloaded than the shell, so the gap can still close; the September 2025 mean can also just have been high. Dodge Momentum Index commercial planning fell in June 2026 on a dip in data-centre filings, and Wood Mackenzie pipeline additions have fallen for three quarters from a 2025Q3 peak. Use \$370–460 bn as a 2026 working range.

The 2× path to \$921 bn is dropped below. By 2027 the paper is simulating new plans, not reading the current stock.

4 Investment intensity

From the 2025 BEA levels, at 2% real,

$$\begin{aligned} I^{GPDI}_{2030} &= 5.46 \times 1.02^5 = \$6.03\,\text{tn}, \quad I^{GPDI}_{2035} = \$6.65\,\text{tn}, \\ I^{PNFI}_{2030} &= 4.25 \times 1.02^5 = \$4.69\,\text{tn}, \quad I^{PNFI}_{2035} = \$5.18\,\text{tn}. \end{aligned} \tag{2}$$

Table 2: Data-centre share of US investment

	I^{DC} (\$bn)	% GPDI	% PNFI 2030	% PNFI 2035
2023 Q4 (Fed)	10	0.2	—	—
2025 Q4 (Fed)	179	3.3	4.2 (2025)	—
2026, 1× inflow	464	8.5 of 2025	—	—
Census × 10/3, Jul 2026	≈ 250	4.6	5.9	—
Low (hold 2027 0.25×)	357	5.9 / 5.4	7.6	6.9
Central (hold 2026 1×)	464	7.7 / 7.0	9.9	9.0
High (hold 2027 1×)	612	10.2 / 9.2	13.0	11.8

The 2023 and 2025 GPDI shares use \$5.46 tn. In the low / central / high rows the GPDI pair is 2030 then 2035.

The central hold is about 8% of GPDI and 10% of PNFI in 2030, from 0.2% of GPDI in 2023. Pre-boom is 2023, not the 2024Q4 Fed print of \$60 bn. Spain is 2.5–3.5% of GFCF in the companion note, so the US intensity is three to four times larger.

Eight percent of GPDI is about 1.4% of GDP if $I/Y \approx 0.18$. Only 44% of I^{DC} is domestic, so the GDP impulse is closer to 0.6% of GDP. The Fed's own 2026 contribution, import-adjusted, is 0.33–0.49 percentage points of nominal GDP growth (Table 2). Equipment depreciates fast, so a 10% PNFI flow share is not a 10% capital-stock share.

5 Electricity intensity

LBNL 2025 Update: 649 TWh in 2030 is 11.8% of the NERC 2025 LTRA forecast, not of 2023 EIA end use. That implies a 2030 total of about 5,500 TWh. Non-data-centre load rises 926 TWh (24%) from 2024 to 2030. Data centres are 33% of load growth over those six years.

Table 3: Electricity share

	E^{DC} (TWh)	Share of E	$\Delta E^{DC}/E_{2023}$
2023	176	4.4%	—
2024	192	4.7%	+0.4%
2030, LBNL low	521	9.5%	+8.6%
2030, LBNL central	649	11.8%	+11.8%
2030, LBNL high	843	15.3%	+16.6%
2035, hold 2030 share	$\approx$ 680	11.8%	$\approx$ +12%
2035, share-drift high	$\approx$ 980	$\approx$ 17%	$\approx$ +20%

In the central 2030 case the data-centre share goes from 4.4% to 11.8%. That ratio does not need an elasticity.

6 Supply elasticity

Electricity supply is layered, and the layers differ.

With $\varepsilon_S = 1$, equation (1) maps a 12% demand shift into a 12% national price. That describes a medium-run, national, annual energy market: other demand frozen, new generation available at rising cost.

There is no clean US estimate of the elasticity of delivered kWh at data-centre nodes. By layer:

- Short-run operating and peak. Wholesale demand elasticities sit near 0–0.35. Existing wind's operating elasticity is about 0.1. At the peak the supply curve is the merit order plus a vertical capacity constraint; PJM capacity prices going from \$29/MW-day to the cap are that market. Use $\varepsilon_S \approx 0.3$ or below.

- Medium-run generation, by fuel. BOEM MarketSim elasticities, from EIA AEO price cases: 1.50 (gas), 0.27 (coal), 0.53 (nuclear), 2.03 (solar), 0.65 (onshore wind), 0.68 (other non-hydro renewables). Weighted by the 2024 US mix (gas 43%, nuclear 18%, coal 15%, wind 10.5%, utility solar 5%), that is about 0.9–1.1. Unit-elastic is a fair medium-run national guess.

- Long-run renewable capacity. A Georgia Tech IV using state RPS schedules puts the price elasticity of renewable capacity near 2.7. That is capacity entry, not kWh at a congested node. It is the right order of magnitude once generation can adjust and transmission is not the bind.

- This shock, in existing models. IMF WP/25/81 (April 2025) has US electricity prices +0.9% under current policies and +8.6% when renewable expansion and transmission are constrained. If the quantity shock is LBNL-central 12%, the implied elasticities are about 13 and 1.4. CMU/NC State's +8% on US generation cost by 2030 is the constrained case again.

The bind for data centres is interconnection, transformers (two to three years) and substations (four to six years), not US generating-fuel supply. The IMF constrained case is the national benchmark; local peak markets can print a multiple of that.

Table 4: National price map, LBNL-central 12% quantity shift

Layer	ε_S	$\Delta P/P$	Reading
Short-run / local peak	0.3	+39%	Merit order plus capacity; PJM-like
Medium-run national generation	1.0	+12%	AEO-weighted fuel mix
IMF constrained T&D	≈ 1.4	+8.6%	Model output
Long-run renewables	2.5	+5%	Capacity entry if the wires exist
IMF current policies	≈ 13	+0.9%	Generation and T&D keep up

For a national number, use about 10% by 2030 when wires and renewables lag: the IMF constrained case is +8.6%, the unit-elastic map is +12%. If generation and wires keep up, about +1%. In the local peak markets the energy session will discuss, several times the national figure. Keep $\varepsilon_S = 1$ for the national annual market; use 0.3 or below for PJM.

7 Funds-market absorption

Split the \$460 bn of investment into retained earnings spent and funds raised.

$$I_t^{DC} = \underbrace{RE_t}_{\text{retained earnings, cash and bills run down}} + \underbrace{F_t}_{\text{bonds, private credit, bank loans, ABS/CMBS}} . \tag{3}$$

r^* and crowding-out respond to ΔI^{DC}. Credit spreads respond to F. From Fed 2023Q4 (\$10 bn) to the central 2026 hold (\$464 bn),

$$\Delta I^{DC} \approx \$450\,\text{bn per year.} \tag{4}$$

The BoE FSR has private credit at 34% of AI-investment financing in 2025, from 9% in 2024. Hyperscalers issued about \$121 bn of US bonds in 2025 (four times the 2020–24 average). Data-centre debt issuance was about \$182 bn. At a 30–50% external share,

$$F \approx \$140\text{–}230\,\text{bn per year} \tag{5}$$

at the central rate. Vontobel's \$300 bn of AI and data-centre bonds over the next year, and \$1.5 tn over five years, is the bond slice only. Barclays forecasts \$945 bn of net US corporate issuance in 2026, so \$300 bn is again about a third of the net calendar.

Table 5: Absorbed funds against the markets that price them

Market M_j	Comparison	Size
US GPDI / saving, \$5.46 tn	$\Delta I^{DC} =$ \$450 bn	8% of the US saving flow
Global saving, order of \$25 tn	$\Delta I^{DC} =$ \$450 bn	about 2% of global saving
Net USD IG issuance	AI-related share, 2025	about 30% of the net calendar; about \$300 bn expected next year
Hyperscaler + DC debt	\$121 bn and \$182 bn in 2025	large single-sector bid
Private credit stock	>\$200 bn AI-related of about \$2.5 tn (BIS)	about 8% of the stock; more than a third of 2025 deals (FSB). BIS projects \$300–600 bn by 2030
Bank C&I to data centres	\$14.9 bn through 2025Q3 (MSCI)	negligible
AI-adjacent bank C&I	\$450 bn commitments, \$150 bn drawn	0.8% of bank assets (Chicago Fed)
Broader AI-company debt	about \$1.2 tn (JPM); \$243 bn of AI-firm bonds in 2025 vs \$3.0 tn NFC issuance (BIS/OECD)	large for a sector; small against all corporate debt
DC ABS/CMBS outstanding	about \$50–60 bn	well measured; small

On the FSB's \$1.5–2.0 tn denominator this would be 10–13% of private credit. BIS later puts outstanding private credit near \$2.5 tn, and AI-related loans at 8% of that stock. More than a third of 2025 originations is the tighter comparison.

$$\frac{\Delta r}{r} \simeq \frac{1}{\eta} \frac{\Delta I^{DC}}{S}. \tag{6}$$

At $\eta = 1$ and $S =$ \$5.46 tn, \$450 bn of extra investment demand raises r by 8% of itself. At a 4% real rate, about 30 basis points. At $\eta = 0.5$, about 65 basis points. Against global saving of order \$25 tn, about 7 basis points at $\eta = 1$.

Those are r^* increments, inside the usual r^* standard error. Put 10–30 bp in a natural-rate discussion. Treasury yields have a much larger denominator.

Thirty percent of net dollar IG issuance is already a thick calendar. Private credit already prices this (8% of the stock, a third of 2025 deals). Banks barely hold it.

Spending retained earnings still uses national saving: cash spent on GPUs is cash not held in bills. What changes is who holds the risk. BIS Bulletin 120 documents capex outgrowing free cash flow. Circular lab–hyperscaler–chipmaker financing inflates gross issuance relative to net absorption, so do not add the \$1.2 tn, \$200 bn and \$182 bn headlines.

8 Working answers

Table 6: Ballparks

Question	Number
Share of US investment, 2030 / 2035	Central: about 8% / 7% of GPDI, 10% / 9% of PNFI. 2023: 0.2% of GPDI.
Share of US electricity vs 2023	4.4% to 11.8% in 2030 (9.5–15.3). One-third of 2024–30 load growth. 2035: about 12% or about 17%.
National electricity price	+9 to +12% by 2030 if wires and renewables lag. About +1% if they keep up. Local peak: a multiple.
Funds taken out	$\Delta I^{DC} \approx \$450$ bn/yr; external $F \approx \$140$–$230$ bn/yr.
r^*?	About 7–30 bp against global / US saving at $\eta = 1$.
Credit rates?	About 30% of net USD IG; about 8% of the private-credit stock, more than a third of 2025 deals. Banks: no.

An adverse scenario that matches these magnitudes is constrained electricity plus private-credit and SPV refinancing.

9 Limitations

The 2030 and 2035 investment numerators hold a 2026–27 Fed rate; they are not in FEDS 2025-109. Wood Mackenzie's 195 GW of signed construction or supply agreements (26% of 2025 US peak load) is a capacity upper bound, not GFCF or TWh. A 30–50% haircut still belongs on utility-stated demand; LBNL is already a haircut relative to those queues.

The price map is partial equilibrium. It leaves out demand destruction, large-load tariffs, behind-the-meter gas (about 2 GW online against 50 GW of IT load), and the split between energy and capacity prices. The generation-weighted elasticity of 0.9–1.1 averages fuels. It is not the elasticity of power at a congested Dominion or ERCOT node.

The funds comparison freezes 2025 financing shares. If hyperscaler free cash flow recovers, F shrinks and the credit-market pressure eases; the r^* arithmetic does not.

Sources

Brandsaas, Garcia, Kurtzman, Nichols and Zytek, "Estimating Aggregate Data Center Investment with Project-level Data," FEDS 2025-109 (Dodge vintage September 2025; Table 2).

Census C30, July 2026 release (1 September 2026): private data-centre VIP $75.166 bn SAAR.

BEA / FRED: GPDI $5.459 tn in 2025 (Table 5.2.5); PNFIA $4.251 tn.

Smith, Hubbard, Newkirk, Ganeshalingam, Holecek, Sartor, Mills and Shehabi, *United States Data Center Energy Usage Report: 2025 Update*, LBNL, June 2026. Denominator: NERC 2025 LTRA.

Bogmans, Gomez-Gonzalez, Ganpurev, Melina, Pescatori and Thube, "Power Hungry: How AI Will Drive Energy Demand," IMF WP/25/81.

BOEM MarketSim own-price supply elasticities (AEO-derived, September 2021 update). Georgia Tech IV estimate of the long-run renewable-capacity elasticity (about 2.7).

BIS Bulletin 120; BIS Bulletin 130; BIS "Financing the digital economy" (2026); BoE FSR (July 2026); FSB private-credit report (May 2026); Vontobel and Barclays on the 2025–26 IG calendar; Chicago Fed (February 2026).